

CANDY DIALOGUE ENGINE

Backgrounds

1.1.0

The following guide provides explanations on how to use Background commands, to control and modify backgrounds during dialogues.

Background Scene

Background commands modify background layers located in a Background Scene.

These Background Scenes are dialogue UI elements. They feature **layer** nodes, which display background images and can be animated. The following nodes can be used as layer nodes: TextureRect, NinePatchRect, TextureButton, Sprite2D/3D, AnimatedSprite2D/3D and VideoStreamPlayer.

The design and layout of Background Scenes is entirely up to you: you can overlay layers, make them occupy the full screen or only part of it, use some layers exclusively for static images and others for moving sprites, mix node types, and so on.

Multiple Background Scenes can be created and swapped at runtime, allowing you to use different background layouts as needed.

Background Commands

The following Background commands are available in Candy DE:

§BGScene

This command swaps the currently loaded Background Scene with another. Simply provide the name of the new scene, and make sure the .tscn file is located in the correct folder. By default:

Candy_DE/Scenes/Background Scenes

To remove a background scene without replacing it with a new one, leave the Scene field empty.

Note: the Default background scene has mouse_filter set to 'stop' on its root node, and the root node occupies the full screen. This means that you won't be able to click through the root node, even if no background layers are displayed.

Therefore, if the background scene overlays game or UI elements that are sometimes meant to respond to mouse interactions, remember to remove the background scene instead of just removing image files from layer nodes.

You should also check the mouse_filter of the default background layer nodes: these nodes are provided for demonstrative purposes and you should normally customize the background scene (or create your own), but if you use the default background scene, you may encounter unexpected behavior when it comes to mouse passing through layer nodes.

§BG

This command assigns or changes an image on a layer node. You need to provide the following data to the command:

Layer → The name of the layer node to assign an image to.

File → The image file name, or the Sprite Frame resource for AnimatedSprite nodes.

Animation → The name of the animation to play (AnimatedSprite nodes only).

Loop → How many times to play the animation.

Wait → Pause dialogue processing until this number of loops have played.

Time → The time or timestamp until which to pause dialogue.

Set 'Loop' to 0 to freeze on the first frame, or to -1 for infinite looping.

Time is written in the HH:MM:SS.00 format, or can be written as frames, like so: f=120.

Time is cumulative with Wait. If the Time (or frames) exceeds the length of an animation, the excess carries over into additional loops.

If Wait/Time exceed the number of Loops, dialogue resumes when all loops have played.

You must create a subfolder for each layer node in Candy_DE/Media/Backgrounds, and each subfolder must be named the same as the layer node it matches. Image or video files are placed inside those subfolders. This means that a layer node named "A" can only ever display the files in the subfolder named "A".

If you use AnimatedSprite nodes, inside each layer subfolder, you must create a folder named "Sprite Frames". Sprite Frame resources must be saved there. For example:

Candy_DE/Media/Backgrounds/A/Sprite Frames

§BGWait

This command pauses dialogue processing until the animations of the designated background layers have played the specified amount of loops and time/frames.

Layers → The layers to wait on, separated with a comma (,).

Wait → Pause dialogue processing until this number of loops have played.

Time → The time amount, timestamp or frame until which to pause dialogue.

See the §BG command above for details on Wait and Time.

§BGStop

This command stops an animation and optionally reset to the first frame or freezes on the current frame. Simply provide the following data:

Layers → The layers to stop animations on, separated with a comma (,).

Default → -1 = Stop on the last frame, 0 = stop on the current frame. 1+ = Pause on the given frame.

§BGRemove

This command clears layers, effectively removing any images, animations or videos.

Layers → The layers to remove images/animations from, separated with a comma (,).

§BGMirror

This command flips a node's image horizontally and/or vertically.

The command only works with nodes that have the flip_h or flip_v properties: TextureRect, TextureButton, Sprite2D, AnimatedSprite2D.

Layers → The layers to mirror, separated with a comma (,).

Axis → Whether to flip horizontally, vertically or both.

§BGEffEffect

This command plays visual effects on layer nodes, in the form of animations through an AnimationPlayer.

Multiple effects can be played simultaneously on the targeted layers. Candy DE will instantiate an AnimationPlayer for each effect and layer, or re-use existing ones if they are unused.

Layers → The layers to play effects on, separated with a comma (,).

Library → The animation library to load.

Effects → The effects to play.

Loop → How many times to play the animation.

Wait → Pause dialogue processing until this number of loops have played.

Time → The time amount, timestamp or frame until which to pause dialogue.

See the §BG command above for details on Loop, Wait and Time.

Animation Library files must be placed in the **Media/Characters/Busts/Animation_Libraries** folder.

§BGEffEffectWait

This command pauses dialogue processing until the designated effects, for the designated background layers, have played the specified amount of loops and time/frames.

Layers → The layers to wait for effects on, separated with a comma (,).

Loop → How many times to play the effect animation.

Wait → Pause dialogue processing until this number of loops have played.

Time → The time amount, timestamp or frame until which to pause dialogue.

See the §BG command above for details on Loop, Wait and Time.

§BGEffEffectStop

This command stops the specified effects for the specified layers.

Layers → The layers to stop effects on, separated with a comma (,).

Effects → The specific effects to stop. If empty, all effects will stop.

Static Backgrounds

If you only plan to use static images for background layers, you can display them inside TextureRect, NinePatchRect or TextureButton nodes.

Simply add the image files to each layer's folder. For example for a layer named 'Layer_1':

Your Project/Candy_DE/Media/General/Backgrounds/Layer_1

You can also display static layers in Sprite2D/3D, AnimatedSprite2D/3D or VideoStreamPlayer nodes: just set the Loop value in the §BG command to 0, and it will pause on the first frame. See the 'Background Animations' section below for details on these nodes.

TextureButton

TextureButton nodes can accept multiple textures, in addition to texture_normal: texture_pressed, texture_hover, etc. Candy DE is designed to handle this automatically if you have corresponding files:

If the layer display node is a TextureButton, a subfolder named "Button Textures" can be created inside the layer's folder. Inside that folder, you can create a subfolder for each normal texture file: name these subfolders exactly as the normal texture file they match to. Use underscore (_) instead of a coma (,) to separate the file extension in the subfolder name.

For example, if the file you plan to assign to texture_normal is named "Clouds.png", name the subfolder "Clouds_png".

In each subfolder, you can add files for the various texture types that TextureButton nodes support: texture_pressed, texture_hover, texture_disabled, texture_focused and texture_click_mask.

Name these files anything you like, but add the texture type as a prefix. For example: "texture_pressed.png" or "Clouds_texture_pressed.png" will both work.

Candy DE will automatically assign the files to the TextureButton node if they exist (having files for all texture types is not required).

To be clear: the file for texture_normal must be placed in the layer's folder (e.g. Candy_DE/Media/General/Backgrounds/Layer_1). It does not need the texture_normal suffix in its name (e.g. "Clouds .png" is fine). This allows you to re-use the same file for other node types.

Background Animations

If you want background layers to play animations, you have three options: Sprite nodes, AnimatedSprite nodes, or VideoStreamPlayer nodes.

For the first two, you should specify the play speed (as frames per second) in **Backgrounds.gd**:

```
@export var sprite_fps = 30.0
```

There is no option to set a custom frame rate for individual animations, except to change the **sprite_fps** variable at runtime with a §Set command before using the §BG command. Therefore, it's recommended that you design all animations for the same frame rate.

Sprite2D/3D

Sprite nodes require a sprite sheet in order to play a bust animation. This is standard: it should be a single image file, split in a grid, where each cell represents an animation frame.

The file name must end with a special tag that will tell Candy DE how many rows and columns the sprite sheet has. For example:

Clouds_4x6-2.png

The tag is the blue part: '_4x6-2' (or any other numbers): 4 is the number of rows (H Frames), 6 is the number of columns (V Frames), and 2 is the number of 'missing' (empty) frames in the last row. If no frames are missing, writing '-0' is optional.

The file format (.png) can be anything you want as long as it's supported by Sprite nodes, and "Clouds" can be replaced with any text you want (e.g. "Frown", "Jump", etc.).

This means you'll need to take care to add the tag in sprite sheet names when creating and saving them, but no extra work will be required beyond that to make Candy DE use them correctly.

AnimatedSprite2D/3D

If you use AnimatedSprite nodes, you'll need to save SpriteFrames resources (.tres).

Unlike other bust files, those SpriteFrames resources must be placed in their own subfolder, named "Sprite Frames", and located inside a layer's own folder. For example:

Your Project/Candy_DE/Media/General/Backgrounds/Layer_1/Sprite Frames

You can create as many SpriteFrames resources as you like for each character, although in most cases one per character is enough.

Make sure your SpriteFrames resources have the correct animations configured, and Candy DE will display them properly if you correctly provide the §BG command with a SpriteFrames resource file name, and the name of the animation track to play.

VideoStreamPlayer

You can use VideoStreamPlayer nodes to display layers. Simply save the video files inside the corresponding layer's folder, then provide the needed data to the §BG command similarly to other node types.

Animated Folders

Note: Animated Folders only work with TextureRect and NinePatchRect nodes. TextureButton nodes are not supported at this time.

You have the option of creating animated background layers from multiple image files, which act as frames for the animation. To do this, follow these steps:

1. Inside a layer's folder, create a folder called "Animated". For example:
Candy_DE/Media/General/Backgrounds/Layer_1/Animated
2. In the Animated folder, create a folder for each animation. Give these folders any name you want.
3. Place the image files for each animation inside its corresponding folder.
4. The files will be displayed in alphabetical order, so name them appropriately (e.g. Smiling_001, Smiling_002, etc.).

Make sure to use zero-padding for numbers: this means writing '001' instead of just '1'. Otherwise, '10' will come before '2' or '9' in alphabetical order.

Be careful with animation folder names that match standalone image or video files: when you provide a file name without an extension to a command (e.g. "Cloud" instead of "Cloud.png"), Candy DE checks if an animation folder of that name exists. If so, it uses that animation.

If no animation folder of the same name exists, Candy DE checks if a file of the same name, with the default extension, exists (e.g. "Cloud.png" if .png is set as the default).

In other words, if you have an animation folder named "Cloud", you can't provide "Cloud" as a substitute for "Cloud.png": Candy DE will use the "Cloud" animation.

Optionally, in each animation folder, you can create a txt file named 'config.txt'. In that file, add this line:

fps = 2.0

This tells the dialogue engine the speed to play each animation at, in frames per second. For example, a value of 5 means that 5 images will be displayed per second.

You can replace the fps value with any number you want, including decimal numbers.

If an animation doesn't have a config.txt file, the engine will use the default **image_fps** value of the portrait box (this is an exported variable in the **Backgrounds.gd** script; you can modify it in the inspector, in the portrait box scene: **Candy_DE/Scenes/Background Scenes/Default.tscn**).

Finally, to make Background layers display an animation, in the \$Background command provide the name of the animation's folder as the file to assign to a Background layer.